

LLM Quality Evaluation

Contract contradiction detection in a waterproofing specification

Task 1: Extraction

Task 2: Evaluation

Best F1

0.89

What I built

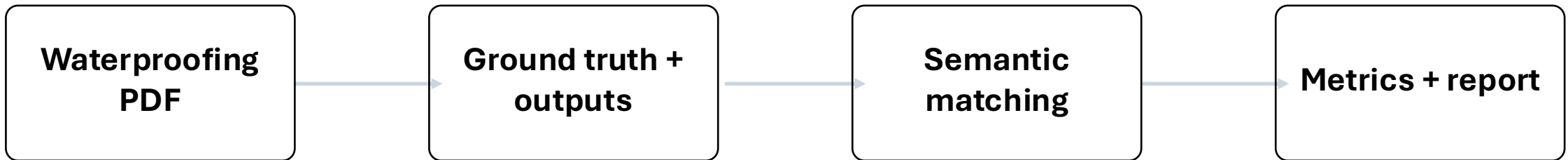
A local Python workflow that extracts model outputs, compares them to ground truth, and reports evaluation metrics such as TP, FP, FN, Precision, Recall and F1.

Key technical choices used in this project

1. Single pass LLM approach
2. Embedding model : all-minilm-l6-v2
3. Matching method: cosine similarity
4. Library: SentenceTransformer
5. Similarity threshold = 0.53
6. LLM Model_Name = gpt-5.6-terra

Assignment

Compare model predictions against a known answer key.



Ground truth is the answer key: 10 known contradictions. Model outputs are model predictions. Task 2 asks which output is safer to trust.

Task 1: single-pass extraction

One LLM call reads the full PDF text and returns structured JSON.

Pipeline

- 1 Extract PDF text with pypdf
- 2 Send full text to OpenAI model
- 3 Ask for Technical, Responsibility, Liability and Schedule contradictions
- 4 Save as my_model_output.json

Output format

```
{  
  "Statement1": "...",  
  "Statement2": "...",  
  "Reasoning": "..."  
}
```

Simple, reproducible, and useful as a baseline. Limitation: the model reads everything at once, so it can still miss subtle conflicts.

Task 2: semantic matching

The same contradiction can be written in different words.

Ground truth

“Membrane minimum thickness is 4 mm,
but another section requires 5 mm.”



Model prediction

“Section 3 allows 4 mm, while Section 7
mandates 5 mm.”

Match rule: combine Statement1 + Statement2 + Reasoning, convert to embeddings, then compare using cosine similarity. If score ≥ 0.53 , count as a match.

How TP, FP and FN are counted

This is the core of the benchmark.

TP True Positive

The model found a real contradiction.

FP False Positive

The model raised a false alarm.

FN False Negative

A real contradiction was missed.

Metrics used

Precision = $TP / (TP + FP)$ Recall = $TP / (TP + FN)$ F1 = balance between Precision and Recall

Evaluation results

My Task 1 output gives the cleanest predictions on this benchmark.

Model	TP	FP	FN	Precision	Recall	F1
My Task 1	8	0	2	1.00	0.80	0.89
Model 1	8	2	2	0.80	0.80	0.80
Model 2	6	4	4	0.60	0.60	0.60

Interpretation

My Task 1 and Model 1 both found 8 real contradictions. My Task 1 produced no false positives, so its precision and F1 are higher.

Visual comparison

F1 score summarizes the balance between false alarms and missed contradictions.



Safer here means: catches many real issues, while creating fewer false alarms for reviewers.

Summary

Focus on method, trade-offs, and honest limits.

Python skills shown

PDF reading, JSON parsing, functions, error handling, embeddings, similarity matrices, metrics, report files.

Method choice

Exact matching is too strict, so I used semantic similarity and a fixed threshold for fair comparison.

Limitations

Only one document and 10 contradictions. The threshold should be validated on more reviewed documents.